Case study brief

Early default model for personal loans

From	Head of Credit Risk
To	Data and Analytics team
Subject	Early default model for unsecured personal loans
Scenario	Fictional South African consumer lender. All data is simulated (Nov 2024 to Oct 2026).
Due	Three weeks from kickoff

Background

We lend small unsecured personal loans, sourced through our app, our branches and 12 referral partners. Too much of our loss comes from loans that go bad in the first six months. Our approval rule is a bureau score cut-off that has not been reviewed in two years: approve if the bureau score is at least 600, decline if the applicant has no bureau record.

The request

Build something that estimates, at the point of application, how likely a customer is to default in the first six months. Tell us whether it beats the current cut-off, whether it treats customers fairly, and how we will know when it stops working.

What we need back

- A short note on anything in the cleaned tables (built by the data pipeline project) that affects the model, such as the labelling window or missing values.
- A default definition we can agree on, and a model that predicts it using only information known at application.
- A like-for-like comparison with the current cut-off, using loans the model has not seen.
- A fairness check by sex and by province.
- A monitoring approach that tells us when the applicant population has moved.
- A short written recommendation (two pages) that a non-specialist can act on, including the limitations.

Data provided

The raw files below are loaded and cleaned by the data pipeline project (bronze, silver and gold tables on Databricks). You start from the gold loan table, one row per loan, built from them.

File	One row per	Notes
customers.xlsx (2 sheets)	customer	Income, employment, bureau score (NH = no hit), province, sex. Two sheets: 2024/25 and 2026.
applications.csv	application	Includes declined applications, partner, decision.
loans.csv	approved loan	Principal, term, interest rate, disbursement date.
repayments.csv	instalment due	Amount due and paid, days past due. Known only after the loan starts.
partners.csv	referral partner	Channel type and commission.

The files come from several systems and are not clean. Expect mixed date and ID formats, duplicated rows, currency stored as text, placeholder values and missing records. There is no default column.

Constraints

- Personal information (ID numbers, names, cell numbers) must be minimised: keep only what the model needs, such as birth year, and drop the rest (POPIA).
- Sex and province must not be used as model inputs. Use them only to check fairness.
- Do not use any repayment information as a model input; it is not known at application.
- Use only the platform we have: Databricks, Python, SQL and PySpark.

Out of scope

Pricing, collections strategy, and estimating how declined applicants would have performed (reject inference).

Success looks like

- We can reproduce your numbers by running your notebooks end to end.
- We can see, in plain language, how the model compares with today's rule at the same approval rate.
- You tell us honestly where the evidence is weak.

Questions to settle at kickoff

Question	Working assumption used in this case study
What counts as default?	90 or more days past due within the first 6 instalments (or fewer for shorter loans).
Which loans can be labelled?	Only loans observed for the full window; newer loans are left out of training and testing.
Which customers are in scope?	Approved and disbursed loans, because outcomes are not observed for declined applicants.
How do we test it?	Train on applications before 1 Jan 2026; test on applications from 1 Jan 2026 onwards.
What does "beats the cut-off" mean?	Compare at the same approval rate (who defaults?) and at the same default rate (how many could be approved?).